

Ilya Emelyanov

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RESEARCH INTERESTS

LLM reasoning, gradient boosting, robust ML for small-sample data, reinforcement learning, applied deep learning.

PUBLICATIONS

Emelyanov, I. (2026). *SmallGBM: Gradient Boosting with Robust Leaf Regularization for Small-Sample Tabular Data*. Zenodo. DOI: [10.5281/zenodo.21934674](https://doi.org/10.5281/zenodo.21934674)

Emelyanov, I. (2026). *SmallMLP: Learned-Bandwidth Nadaraya-Watson Regression with Prediction Intervals for Small Nonlinear Data*. Preprint in preparation.

RESEARCH EXPERIENCE

- SmallGBM** — Gradient Boosting for Small-Sample Tabular Data github.com/nsdmlk/SmallGBM · PyPI: [smallgbm](#)
- Designed and implemented a gradient boosting library for $n < 1000$
 - Novel contributions: robust leaf weight estimation (median for small leaves, inverse-distance weighted mean for larger leaves), signal-adaptive shrinkage toward parent nodes, stochastic split selection
 - Benchmarked on 27 datasets (15 synthetic + 12 real-world): AUC 0.9241 vs. 0.9156 (XGBoost), 0.9047 (LightGBM), with lowest variance (± 0.0676)
 - Published on PyPI; scikit-learn compatible API
- SmallMLP** — Learned-Bandwidth Nadaraya-Watson Regression for Small Nonlinear Data github.com/nsdmlk/SmallMLP · PyPI: [smallmlp](#)
- Designed a non-parametric regressor for small, nonlinear data ($n < 500$): per-feature kernel bandwidth learned via leave-one-out optimization, removing manual tuning
 - Added weighted conformal prediction intervals with finite-sample coverage guarantees, adapting interval width to local data density
 - Outperformed standard MLPs on 45 benchmark datasets (mean rank 1.80, 25/45 wins); conformal intervals 19% narrower than split conformal at equal coverage
- LLM-Core** — Transformer from Scratch github.com/nsdmlk/LLM-Core
- Implemented Multi-Head Attention, encoder-decoder architecture, and BPE tokenizer without pre-built model libraries
 - Trained on a 104M-word technical text corpus
- GraphFlow** — Graph-Based Road Network Analysis (in progress) github.com/nsdmlk/GraphFlow
- Graph algorithms (Dijkstra, A*) combined with ML-based traffic prediction

ML ENGINEERING

- SberAutoPodpiska** — Conversion Prediction Service github.com/nsdmlk/sberautopodpiska-ml
- Binary classification on 1.86M user sessions, 13 features, ROC-AUC 0.706
 - Deployed as a FastAPI REST service, containerized with Docker
- Rubium** — Open-Source Educational Platform github.com/qqwzz/Rubium · [Apache 2.0](#)
- Lead Developer (3-person team): architecture, backend, coordination
 - Stack: Vue 3, Go (Gin), Python, Supabase, TipTap

EDUCATION

Beijing Institute of Technology — B.S. Computer Science and Technology · Full International Scholarship Sep 2026 – Jun 2030

Harbin Institute of Technology — Preparatory Programme Aug 2025 – Jun 2026

SKILLS

ML PyTorch, scikit-learn, LightGBM, XGBoost, Hugging Face, Pandas, NumPy

Backend Python, Go (Gin), Java, FastAPI, Flask

Infra Docker, Git, Linux, SQL (PostgreSQL, MySQL, Supabase)

Math Linear Algebra (MIT 18.06), Probability, Statistics

LANGUAGES

Russian — Native | English — B2–C1 | Chinese — HSK4